Dynamic Programming

Part 1: Exercise 6, 7 & 8 (week 9)

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Plan for today

- Continuous shocks
- Deaton's model: A more realistic consumption-saving model
- Euler Equation and Euler errors
- Advanced interpolation
- Exercise 6, 7 & 8

Exercise 6 + 7: Deaton's model

Much like cate-eating but with benefit of saving, CRRA-utility and log-normal income. New **Bellman equation**:

$$V_t(W_t) = \max_{C_t} \left\{ \frac{C_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}[V_{t+1}(W_{t+1})] \right\}$$

s.t.

$$W_{t+1} = R(W_t - C_t) + Y_{t+1}$$

$$Y_{t+1} = \exp(\xi_{t+1})$$

$$\xi_{t+1} \sim \mathcal{N}(0, \sigma_\xi^2)$$

$$A_t = W_t - C_t \geq 0$$

for finite horizon.. drop time-subcripts $\rightarrow$ Infinite horizon

Exercise 6 + 7: Gauss-Hermite for log-normal r.v.

Consider $X \sim \exp\left(\mathcal{N}(\mu, \sigma^2)\right)$ with $supp(X) = \mathbb{R}_+$, some function $f(\cdot)$ and compute expectation:

$$\mathbb{E}[f(X)] = \int_0^\infty f(x) \overbrace{\frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln(x)-\mu)^2}{2\sigma^2}\right)}^{\text{log-normal r.v. PDF}} dx$$

Change variables $z = \frac{\ln(x)-\mu}{\sqrt{2\sigma^2}} \Rightarrow dz = \frac{1}{x\sqrt{2\sigma^2}} dx \Leftrightarrow dx = x\sqrt{2\sigma^2} dz$ and rewrite to "Gauss-Hermite form", using $x = \exp\left(\mu + \sqrt{2\sigma^2} z\right)$

$$\Rightarrow \mathbb{E}[f(X)] = \frac{1}{\sqrt{\pi}} \int f\left(\exp\left(\mu + \sqrt{2\sigma^2} z\right)\right) \exp(-z^2) dz$$

$$\approx \frac{1}{\sqrt{\pi}} \sum_{i=1}^{n} w_i f\left(\exp\left(\mu + \sqrt{2\sigma^2} z\right)\right)$$

Only change compared to normal r.v.

Exercise 6 + 7: Euler errors

How do we evaluate how well we solved the model?

$\rightarrow$ Use optimality condition: the **Euler Equation**.

For simulated households $i = 1, 2, \ldots, N$:

$$u'(C_{i,t}) = \beta R \mathbb{E}_t[u'(C_{i,t+1})]$$

How well does the solution fulfill condition that should be fulfilled?

Define mean Euler error:

$$\frac{1}{\sum_{i=1}^{N} \sum_{t=1}^{T-1} 1_{\{A_{i,t}>0\}}} \sum_{i=1}^{N} \sum_{t=1}^{T-1} 1_{\{A_{i,t}>0\}} \left| u'(C_{i,t}) - \beta R \mathbb{E}_t[u'(C^*_{i,t+1})] \right|$$

, where we only consider unconstrained households, compute expectations with quadrature, evaluate next-period policy $C^*_{i,t+1}$ and only consider periods $t = 1, 2, \ldots, T-1$

Deriving the Euler Equation 1/2. Bellman step

Use Bellman equation + The Envelope Theorem

$$V_t(W_t) = \max_{C_t} \left\{ \frac{C_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}[V_{t+1}(W_{t+1})] \right\}$$

$$\text{s.t.} \quad W_{t+1} = R(W_t - C_t) + Y_{t+1}$$

Dif' within the max-operator and add * (for optimality condition)

$$C_t^{*-\rho} + \frac{\partial \beta \mathbb{E}[V_{t+1}(W_{t+1})]}{\partial W_{t+1}} \overbrace{\frac{\partial W_{t+1}}{\partial C_t}}^{=-R} = 0$$

$$\Leftrightarrow C_t^*(W_t)^{-\rho} = \beta R \frac{\partial \mathbb{E}[V_{t+1}(W_{t+1})]}{\partial W_{t+1}} = \beta R \mathbb{E}\left[\frac{\partial V_{t+1}(W_{t+1})}{\partial W_{t+1}}\right]$$

Bellman eq. w/o max-operator, instead optimal policy $C_t^*(W_t)$

$$V_t(W_t) = \frac{C_t^*(W_t)^{1-\rho}}{1-\rho} + \beta \mathbb{E}[V_{t+1}(W_{t+1})]$$

$$\text{s.t.} \quad W_{t+1} = R(W_t - C_t^*(W_t)) + Y_{t+1}$$

Deriving the Euler Equation 2/2. Envelope step

Next, compute $\frac{\partial V_t(W_t)}{\partial W_t}$. Two "effects" pop up: indirect and direct

$$\frac{\partial V_t(W_t)}{\partial W_t} = \overbrace{C_t^*(W_t)^{-\rho}\frac{\partial C_t^*}{W_t} + \frac{\partial \beta \mathbb{E}[V_{t+1}(W_{t+1})]}{\partial W_{t+1}}\frac{\partial W_{t+1}}{\partial C_t^*}\frac{\partial C_t^*}{\partial W_t}}^{=0}$$

$$+\frac{\partial \beta \mathbb{E}[V_{t+1}(W_{t+1})]}{\partial W_{t+1}}\overbrace{\frac{\partial W_{t+1}}{\partial W_t}}^{=R}$$

$$= \underbrace{\beta R\frac{\partial \mathbb{E}[V_{t+1}(W_{t+1})]}{\partial W_{t+1}}}_{\text{From dif' of max: } = C_t^*(W_t)^{-\rho}} = C_t^*(W_t)^{-\rho}$$

$$\Rightarrow \frac{\partial V_{t+1}(W_{t+1})}{\partial W_{t+1}} = C_{t+1}^*(W_{t+1})^{-\rho}$$

Envelope theorem: indirect effect $= 0$

Last step: Insert $\frac{\partial V_{t+1}(W_{t+1})}{\partial W_{t+1}} = C_{t+1}^*(W_{t+1})^{-\rho}$ and get:

$$\Rightarrow C_t^*(W_t)^{-\rho} = \beta R\mathbb{E}\left[C_{t+1}^*(W_{t+1})^{-\rho}\right]$$

Exercise 8: Function Approximation

Exercise 3: linear ("spline") interpolation in 1D

Many other alternatives: Here for $n+1$ known points $\{(x_i, y_i)\}_{i=1}^{n+1}$

- Cubic splines: Fit n 3rd degree polynomials $\{C_i(x)\}_{i=1}^{n}$ by demanding 1) $C_i(x_{i-1}) = y_{i-1}$ and $C_i(x_i) = y_i$ for $i = 1, \ldots, n$
  2) $C_i'(x_i) = C_{i+1}'(x_i)$ and $C_i''(x_i) = C_{i+1}''(x_i)$ for $i = 1, \ldots, n-1$
  3) 2 boundary conditions
- Higher-order polynomial regression: Make polynomial transformations of $\{x_i\}_{i=1}^{n+1}$ and run OLS
- Chebyshev polynomial regression

Caution $\rightarrow$ Runge's phenomenon: Polynomials can *go crazy* between nodes

Your time to shine!